

GeoIntel AI — complete learning syllabus

A full checklist of everything to understand across all five pillars. No timeline attached — pull items as the build needs them, and use this to check you haven't missed anything.

1. Software engineering

1.1 Python core

- ☐ Data types, control flow, functions, classes (OOP refresher coming from Java)
- ☐ Virtual environments and dependency management (venv, poetry, or uv)
- ☐ Type hints and static checking (mypy basics)
- ☐ `async`/`await` syntax and asyncio fundamentals
- ☐ Context managers (`with` statements)
- ☐ Decorators (used throughout FastAPI)
- ☐ Custom exceptions and error handling patterns
- ☐ Working with JSON serialization/deserialization
- ☐ Datetime and timezone handling (critical for cross-source news timestamps)
- ☐ Regular expressions for text cleaning and keyword filtering
- ☐ Package/module structure and import conventions

1.2 FastAPI

- ☐ Path parameters, query parameters, request bodies
- ☐ Pydantic models for request/response validation
- ☐ Dependency injection (`Depends()`)
- ☐ Async endpoints vs sync endpoints — when each applies
- ☐ Background tasks vs external schedulers
- ☐ Middleware (CORS, logging, custom auth)
- ☐ Exception handlers
- ☐ Auto-generated OpenAPI/Swagger docs
- ☐ Testing with `TestClient` and pytest
- ☐ Cookie/session-based authentication

1.3 PostgreSQL

- ☐ Relational schema design: tables, keys, foreign keys, indexes
- ☐ SQL fundamentals: SELECT, JOIN, GROUP BY, subqueries, window functions
- ☐ Migrations and schema versioning (Alembic)
- ☐ Transactions and ACID basics
- ☐ `pgvector` extension: vector columns, similarity operators
- ☐ Vector indexing at scale (IVFFlat, HNSW) — conceptual, for later

- ☐ Connection pooling (asyncpg / SQLAlchemy async engine)
- ☐ JSONB columns for semi-structured data
- ☐ Full-text search (`tsvector`) as a complement to semantic search

1.4 Redis

- ☐ Key-value basics and TTLs
- ☐ Cache-aside pattern
- ☐ Rate limiting patterns
- ☐ Pub/sub basics
- ☐ Redis as a Celery broker

1.5 APIs

- ☐ REST principles: idempotency, status codes, resource modeling
- ☐ API versioning strategies
- ☐ Pagination patterns (cursor vs offset)
- ☐ Rate limiting and API key design for your own endpoints
- ☐ Consuming third-party APIs: auth headers, retries, backoff
- ☐ Webhook basics
- ☐ API documentation practices

1.6 Docker

- ☐ Images vs containers
- ☐ Writing a Dockerfile, multi-stage builds
- ☐ Docker Compose for multi-service local dev
- ☐ Volumes and persistent storage concepts
- ☐ Environment variable management in containers
- ☐ Container-to-container networking (service discovery by name)
- ☐ Image size optimization

1.7 Cloud deployment

- ☐ PaaS vs IaaS vs serverless — when each fits
- ☐ Secrets/environment variable management in production
- ☐ CI/CD basics (build/test/deploy pipeline, e.g. GitHub Actions)
- ☐ Domain/DNS basics, HTTPS/TLS certificates
- ☐ Production logging and basic alerting
- ☐ Free-tier limits and scaling cost awareness
- ☐ Horizontal vs vertical scaling concepts

1.8 System design

- ☐ Load balancing basics

- ☐ Where caching layers fit in a request path
- ☐ Message queues and async processing (why ingestion and enrichment are decoupled)
- ☐ Database indexing and query optimization
- ☐ Read replicas / scaling reads vs writes
- ☐ Backpressure and rate limiting under load
- ☐ Idempotency in distributed/background jobs
- ☐ CAP theorem basics
- ☐ Designing for failure: retries, circuit breakers, dead-letter queues

1.9 MLOps

- ☐ Prompt/model versioning — tracking which version produced which output
 - ☐ Evaluation pipelines (your extraction eval set is the seed of this)
 - ☐ Logging LLM inputs/outputs for debugging
 - ☐ Token/cost monitoring per model call
 - ☐ A/B testing prompts or models
 - ☐ Data pipeline health monitoring (did ingestion actually run today?)
 - ☐ Model/prompt drift over time
 - ☐ Basic feature store concepts
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2. Artificial intelligence

2.1 Machine learning foundations

- ☐ Supervised vs unsupervised vs reinforcement learning
- ☐ Train/validation/test splits
- ☐ Overfitting vs underfitting
- ☐ Precision, recall, F1, accuracy (needed for your extraction eval)
- ☐ Classification vs regression, conceptually

2.2 Deep learning foundations

- ☐ Neural network basics: layers, weights, activation functions
- ☐ Backpropagation and gradient descent, intuition-level
- ☐ Regularization concepts
- ☐ Why transformers replaced RNNs/LSTMs for language

2.3 NLP fundamentals

- ☐ Tokenization: word, subword, byte-pair encoding
- ☐ Named entity recognition (you're already using spaCy for this)
- ☐ Text classification and sentiment analysis basics
- ☐ Extractive vs abstractive summarization

- ☐ Stop words, stemming, lemmatization
- ☐ Language detection and multilingual considerations

2.4 Transformers

- ☐ Attention mechanism: query/key/value intuition
- ☐ Self-attention vs cross-attention
- ☐ Encoder vs decoder vs encoder-decoder architectures
- ☐ Positional encoding — why order needs explicit representation
- ☐ Context window limits and their downstream effect on your pipeline
- ☐ Model scaling basics (why bigger models behave differently)

2.5 LLMs

- ☐ Zero-shot, few-shot, chain-of-thought prompting
- ☐ Structured JSON output prompting (already in use)
- ☐ Temperature, top-p, and sampling parameters
- ☐ System prompts vs user prompts
- ☐ Context window budgeting
- ☐ Causes of hallucination and mitigation strategies
- ☐ Cost vs quality vs latency trade-offs across model tiers
- ☐ Function calling / tool use basics
- ☐ Token accounting and cost estimation

2.6 Embeddings

- ☐ What a vector embedding represents semantically
- ☐ Cosine similarity vs dot product vs Euclidean distance
- ☐ Embedding model trade-offs: dimension size, multilingual support, cost
- ☐ Chunking strategies for longer documents
- ☐ Embedding raw text vs a summary — trade-offs
- ☐ Vector index types: flat vs IVFFlat vs HNSW, conceptually

2.7 RAG

- ☐ Where RAG systems actually fail: retrieval vs generation
- ☐ Re-ranking retrieved results (cross-encoder re-ranking, conceptual)
- ☐ Context assembly — how much retrieved content to include
- ☐ Grounding techniques — citing sources, refusing when ungrounded
- ☐ Hybrid search — combining keyword and semantic retrieval
- ☐ Evaluating RAG: retrieval precision/recall, answer faithfulness

2.8 AI agents

- ☐ Single LLM call vs an agentic loop

- ☐ Tool use / function calling as the agent action mechanism
- ☐ Planning and multi-step reasoning patterns
- ☐ LangGraph basics: nodes, edges, state
- ☐ Agent memory: short-term context vs longer-term stored state
- ☐ Failure handling in agent loops: retries, fallback steps, human checkpoints
- ☐ Cost and latency implications of multi-step chains

2.9 Knowledge graphs

- ☐ Entities, relationships, triples — the basic graph data model
- ☐ When a knowledge graph earns its complexity over a relational table
- ☐ Basic graph query concepts (Cypher or a lightweight relational approximation)
- ☐ Entity resolution/deduplication across naming variants
- ☐ Temporal graphs — representing how relationships change over time

2.10 MCP

- ☐ What MCP standardizes for tool/resource exposure to LLMs
 - ☐ Building a simple MCP server for one of your own tools
 - ☐ Connecting an agent to external MCP servers vs your own
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3. Economics & finance

3.1 Macroeconomics

- ☐ GDP and growth-cycle basics
- ☐ Inflation causes and measurement (CPI, WPI)
- ☐ How central banks set interest rates and why
- ☐ Monetary policy vs fiscal policy — who does what
- ☐ Unemployment/inflation relationship, conceptually
- ☐ Balance of trade / current account basics

3.2 Stock markets

- ☐ How exchanges work: order books, market makers, matching
- ☐ Equity valuation basics: P/E ratio, market cap, earnings
- ☐ Index construction (how Nifty/Sensex/S&P 500 are built)
- ☐ Market sentiment indicators (VIX, put-call ratio)
- ☐ Sectoral sensitivity — why energy stocks react to oil events specifically

3.3 Commodities

- ☐ Spot vs futures pricing
- ☐ Crude oil benchmarks: Brent vs WTI vs Dubai, and why they diverge

- ☐ Supply/demand shocks and transmission into price
- ☐ OPEC+ mechanics: quotas, compliance, spare capacity
- ☐ Strategic petroleum reserves as a shock buffer

3.4 Interest rates & bonds

- ☐ Bond basics: yield, coupon, face value, price-yield inverse relationship
- ☐ Yield curve shape and inversion signals
- ☐ Sovereign bonds and country risk premiums
- ☐ How rate changes ripple into currencies and equities

3.5 Inflation

- ☐ Headline vs core inflation
- ☐ Energy price pass-through to broader inflation
- ☐ Real vs nominal returns

3.6 Currencies

- ☐ Floating vs pegged exchange rate mechanics
- ☐ Petrocurrencies — why oil-exporter currencies track oil prices
- ☐ INR-specific dynamics: RBI intervention, import-bill sensitivity to crude
- ☐ Currency as a transmission channel for geopolitical shocks

3.7 Investing fundamentals

- ☐ Risk vs return, diversification basics
 - ☐ Time horizon and position sizing concepts
 - ☐ Trader's reaction window vs long-term investor's horizon
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4. Geopolitics

4.1 International relations foundations

- ☐ Realism vs liberalism vs constructivism, at a conceptual level
- ☐ State actors vs non-state actors (militias, terrorist groups, multinationals)
- ☐ Major current alliances and blocs (NATO, BRICS, Quad, etc.)

4.2 Wars & conflicts

- ☐ Causes and escalation patterns of modern conflicts
- ☐ Market reaction patterns: onset vs prolonged conflict vs resolution
- ☐ Proxy wars and their harder-to-price economic signaling
- ☐ Current conflict zones relevant to your slice (Middle East, Russia-Ukraine)

4.3 Trade

- ☐ Trade agreements and blocs, WTO basics
- ☐ Tariffs and their market impact
- ☐ Supply chain chokepoints: Strait of Hormuz, Suez Canal, Strait of Malacca
- ☐ Export/import bans — immediate vs lagged market effects

4.4 Sanctions

- ☐ Types of sanctions: trade, financial, sectoral, individual
- ☐ How sanctions are enforced and evaded
- ☐ Case studies: Iran, Russia, Venezuela sanctions and market effects
- ☐ SWIFT and financial sanctions mechanics

4.5 Energy politics

- ☐ OPEC+ decision-making and member incentives
- ☐ Energy independence strategy (shale, renewables transition politics)
- ☐ Pipeline politics (Nord Stream and regional pipeline disputes)
- ☐ Strategic reserve releases as a policy tool

4.6 Diplomacy

- ☐ How diplomatic signaling works: statements, summits, back channels
- ☐ Reading escalation vs de-escalation signals in official language
- ☐ Role of international mediators

4.7 Global organizations

- ☐ UN Security Council mechanics: veto powers, resolutions
 - ☐ IMF/World Bank roles in crisis response
 - ☐ IEA — as both a data source and a policy influence
 - ☐ G7/G20 coordination role during shocks
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5. Entrepreneurship

5.1 Product thinking

- ☐ Problem-solution fit vs product-market fit
- ☐ Defining MVP scope (already practiced via the thin-slice approach)
- ☐ Feature prioritization framework — pick one (RICE or MoSCoW), not both

5.2 User experience

- ☐ UX principles for information-dense products
- ☐ Designing for trust: citations, confidence levels, source transparency

- ☐ Mobile-first design principles (already applied in the tracker)

5.3 Startup strategy

- ☐ Competitive positioning against Bloomberg/Perplexity/OpenAI-style products
- ☐ Niche-first strategy — why the oil/energy slice beats going broad early
- ☐ What actually forms a moat in an AI product: data, workflow, trust — not the model itself

5.4 Business models

- ☐ Subscription vs usage-based vs freemium trade-offs
- ☐ Unit economics: cost per query/report vs willingness to pay
- ☐ B2C vs B2B2C (direct to traders vs distribution through a broker)

5.5 Execution

- ☐ Shipping cadence and avoiding scope creep
- ☐ Talking to real users before building more
- ☐ Deciding what not to build

5.6 Decision making

- ☐ Reversible vs irreversible decisions — moving fast on the former only
- ☐ Distinguishing signal from noise in user feedback
- ☐ Recognizing when to pivot vs when to persist on the current slice